Sentiment Analysis Report on the Amazon Reviews Dataset

Report

Introduction

The dataset used is the Datafiniti Amazon Consumer Reviews dataset (obtained from Kaggle). The CSV has 23 columns and 28332 entries. The dataset is a collection of reviews for different products purchased on Amazon and these reviews consist of review text, ratings and product information. The review.text column was selected to be used with the spaCy medium English model as it has the customer's written testimony of their experience with the product. This text would then be used to gain the insights to conduct sentiment analysis.

Data Preprocessing

When looking at the data with .head() and scrolling across the columns in the data, we can see that there are many instances of "NaN". This is indicative of missing data. Any entries with empty review.text columns were removed using dropna(). Fortunately after these rows were removed from the dataframe, there were still 28332 entries (meaning there were no empty review.text fields within the data. The text in the reviews had their leading and trailing spaces removed, the text was converted to lowercase, stop words were removed and punctuation was removed. This was done to improve the effectiveness of the model.

Evaluation of Results

The sentiment analysis model classified each review in one of three ways:

- Positive
- Negative
- Neutral

The reviews also had polarity scores, which ranged from -1 to 1. Positive reviews would generally produce a positive polarity score, and negative reviews would generally produce a negative polarity score.

The reviews could be assigned a similarity score, which was achieved by comparing two reviews and making use of spaCy's word vectors.

Strengths and Limitations

The model shines in that it is easy to implement and uses pretrained language models. It can provide sentiment classification and polarity scores, meaning that it can lend perspective to which degree the customer likes or dislikes the product. The model can also compare reviews and see their similarity score.

The model struggles with regard to a broader and more nuanced understanding of language and choice of words from the customer. Sarcasm, good or bad, may be used by the customer and the model could misinterpret this. This misunderstanding can also extend to context-specific language (jargon, colloquialisms etc.). The sentiment model depends on the context provided by the pretrained rules of the language model.

Distinct language is beneficial to this context. Consider the following sentences:
"I love this TV."
"I hate this TV."

They may have a high similarity score because only 1 word differs between the sentences, but their sentiments differ vastly.

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